

Finite Element Analysis of a Seven Layer Skateboard Deck: Comparative Static Structural Study of Flat and Curved Maple Profiles with Carbon Fiber Composite Material Substitution

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Abstract—This study presents a three stage finite element analysis (FEA) of a complete skateboard assembly carried out in ANSYS Workbench 2025 R2. The purpose of the investigation is to quantify how two common design choices, board curvature and deck material, each influence the structural performance of a production scale skateboard under a realistic impact load. Stage 1 establishes a baseline comparison between a flat and a curved seven ply hard rock maple deck. Both configurations include the full assembly: aluminum alloy 357.0 trucks, rigid TPU wheels, hardened carbon steel hardware, and carbon steel bearing assemblies. A 5,000 N quasi-static load, representative of a two-foot drop, is applied at the deck center while fixed supports are applied at the four wheel contact patches. Each maple ply is assigned orthotropic elastic properties, and cross-ply grain orientation is introduced in selected plies using custom ANSYS coordinate systems. The curved deck reduces maximum total deformation from 0.855 mm (flat) to 0.547 mm, a 36% improvement, and is therefore selected for the second stage. A six-point mesh convergence study confirms solution stabilization between 10 and 18 mm element sizes. In Stage 2 the curved deck is replaced with a seven layer engineered Epoxy Carbon laminate, which reduces maximum deformation further to 0.445 mm, a total 48% improvement relative to the flat maple baseline. FEA results are validated against classical Euler–Bernoulli beam theory, global equilibrium is verified, and factors of safety are computed against the appropriate strength limits for each material system. Model limitations, the incomplete explicit dynamic study, and design recommendations for the truck interface region are discussed. All materials used were taken directly from the built-in ANSYS Engineering Data library to support reproducibility of the analysis.

Keywords: ANSYS Workbench, FEA, skateboard deck, maple, orthotropy, carbon fiber composite, laminate, static structural, mesh convergence, von Mises stress, factor of safety

NOMENCLATURE

$[K]$	Global stiffness matrix
$\{u\}$	Nodal displacement vector
$\{F\}$	Applied force vector (N)
E_{11}	Longitudinal modulus (GPa)
E_{22}	Transverse modulus (GPa)
G_{12}	In-plane shear modulus (GPa)
ν_{12}	In-plane Poisson’s ratio
I	Second moment of area (m ⁴)
M	Bending moment (N·m)
c	Neutral axis distance (m)
F, L	Applied load (N); span length (mm)
b, h	Deck width; total thickness (mm)
δ	Max total deformation (mm)
σ_b	Bending stress (MPa)
σ_{vM}	Von Mises equivalent stress (MPa)
ϵ_{eq}	Equivalent (von Mises) elastic strain
ϵ_1	Maximum principal elastic strain
σ_{ult}	Ultimate strength (MPa)
ρ	Material density (kg/m ³)
θ	Fiber/grain orientation angle (deg)
FoS	Factor of safety
MOR	Modulus of rupture
UD/CFRP	Unidirectional; carbon fiber polymer
TPU/FEA	Thermoplastic polyurethane; FE analysis

1 INTRODUCTION

1.1 Problem Identification

The modern skateboard deck is a laminated composite structure that must satisfy a demanding and conflicting set of engineering requirements. It must be stiff enough to provide precise pop and response, compliant enough to absorb rider impact energy, strong enough to resist fracture under repeated dynamic loading, and light enough to remain maneuverable. A professional-grade deck fabricated from seven plies of hard rock maple (*Acer saccharum*) with cross-plyed grain orientations has served as the industry standard for decades [1]. As rider skill levels and trick amplitudes have increased, however, so have the structural demands placed on the deck.

A rider executing a two-foot (0.61 m) ollie or drop transmits a peak ground reaction force well in excess of 4,000 N to the center of the deck, which is also the mechanically weakest point in the span between the two truck bolt clusters. In this configuration the deck behaves very much like a simply supported beam carrying a central point load, a textbook bending problem that finite element analysis can solve with high fidelity while also revealing the full three-dimensional stress distribution, including local concentration effects at hardware interfaces that a one-dimensional beam model cannot capture.

Carbon fiber reinforced polymer (CFRP) decks have entered the premium skateboard market in recent years, but rigorous comparative data under controlled FEA conditions with realistic full-assembly modeling are sparse in the open literature. The structural role of board curvature, the concave profile fa-

miliar to every skater, is also well understood qualitatively but rarely quantified analytically. Addressing both of these gaps is the motivation for the present study.

A seven-ply hard rock maple laminate was selected as the baseline configuration because it reflects the established industry standard for skateboard deck construction. In this design, all plies are composed of the same wood species (hard rock maple), rather than a combination of different materials, ensuring uniform material behavior and predictable mechanical properties. Structural performance is achieved through the orientation of each ply, where adjacent layers are arranged in alternating grain directions (cross-ply configuration). This directional layering enhances stiffness, strength, and resistance to warping without altering the base material system and is consistent with standard skateboard manufacturing practices [2].

1.2 Objectives

This study uses ANSYS Workbench 2025 R2 to fulfill the following engineering objectives:

1. Build a full-detail SolidWorks assembly (deck, trucks, wheels, bearings, hardware), export to ANSYS via STEP 203, and prepare the geometry in SpaceClaim.
2. Assign realistic orthotropic material models to all seven maple plies using custom coordinate systems for alternating grain direction.
3. Perform a parametric mesh convergence study and document element counts, mesh quality, and convergence behavior.
4. Execute a static structural comparison between a flat and a curved maple deck under a 5,000 N central quasi-static load with fixed wheel supports, reporting total deformation, equivalent elastic strain, maximum principal elastic strain, and von Mises stress.
5. Select the superior deck geometry and repeat the analysis with a seven layer engineered CFRP laminate, demonstrating the material substitution benefits.
6. Validate FEA results against classical beam theory hand calculations, compute the percentage error, and verify global equilibrium.

2 ANALYTICAL SOLUTION

2.1 Governing FEA Equation

All three analyses solve the discretized static equilibrium problem

$$[\mathbf{K}]\{\mathbf{u}\} = \{\mathbf{F}\}, \quad (2.1)$$

where $[\mathbf{K}]$ is the assembled global stiffness matrix, $\{\mathbf{u}\}$ is the unknown nodal displacement vector, and $\{\mathbf{F}\}$ is the externally applied force vector. For orthotropic solid elements, each element's stiffness is built from the constitutive relation

$$\{\sigma\} = [\mathbf{C}]\{\epsilon\}, \quad (2.2)$$

with $[\mathbf{C}]$ being the 6×6 orthotropic elasticity tensor containing nine independent constants ($E_1, E_2, E_3, \nu_{12}, \nu_{13}, \nu_{23}, G_{12}, G_{13}, G_{23}$) that vary per ply due to grain orientation rotation.

2.2 Von Mises Criterion

For isotropic components (hardware, bearings, trucks), the von Mises equivalent stress is the appropriate failure metric:

$$\sigma_{\text{VM}} = \left[\frac{1}{2} ((\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2) \right]^{1/2}, \quad (2.3)$$

where $\sigma_1, \sigma_2, \sigma_3$ are the principal stresses. For the orthotropic wood laminate, maximum principal elastic strain ϵ_1 is reported as the most sensitive indicator of grain-direction failure, since wood is brittle across the grain.

2.3 Impact Load Estimation

To justify the 5,000 N load value, consider a skater of mass $m = 75$ kg (typical adult) performing a two-foot drop from height $h_d = 0.61$ m (24 in). Assuming perfectly rigid ground and a soft-landing contact duration of $\Delta t = 50$ ms:

$$v_{\text{impact}} = \sqrt{2gh_d} = \sqrt{2 \times 9.81 \times 0.61} = 3.46 \text{ m/s}, \quad (2.4)$$

$$F_{\text{avg}} = \frac{mv_{\text{impact}}}{\Delta t} = \frac{75 \times 3.46}{0.05} = 5,190 \text{ N}. \quad (2.5)$$

A design load of 5,000 N is therefore a reasonable conservative quasi-static equivalent.

2.4 Classical Beam Theory Deflection

The skateboard deck is approximated as a simply supported beam of effective span $L = 356$ mm (truck-to-truck axle distance), uniform width $b = 203$ mm, and total laminate thickness $h = 11$ mm. The second moment of area for a rectangular cross section is

$$I = \frac{bh^3}{12} = \frac{0.203 \times (0.011)^3}{12} = 2.25 \times 10^{-8} \text{ m}^4. \quad (2.6)$$

The midspan deflection under a central point load $F = 5,000$ N, using the longitudinal maple modulus $E_{11} = 12.5$ GPa, is

$$\begin{aligned} \delta_{\text{theory}} &= \frac{FL^3}{48E_{11}I} \\ &= \frac{5000 \times (0.356)^3}{48 \times (12.5 \times 10^9)(2.25 \times 10^{-8})} = 0.749 \text{ mm}. \end{aligned} \quad (2.7)$$

2.5 Classical Beam Theory Bending Stress

Maximum bending stress at the outer fiber (tension side, bottom ply) under the same simply supported central load assumption:

$$M_{\text{max}} = \frac{FL}{4} = \frac{5000 \times 0.356}{4} = 445 \text{ N}\cdot\text{m}, \quad (2.8)$$

$$\sigma_b = \frac{M_{\text{max}} c}{I} = \frac{445 \times 0.0055}{2.25 \times 10^{-8}} = 108.8 \text{ MPa}. \quad (2.9)$$

This bending stress will be compared to the material strength to compute the factor of safety in Section 8.

2.6 Equilibrium Verification

For a simply supported beam with four-point support (four wheel contacts), symmetry dictates equal reaction forces. The total applied load must be balanced:

$$\sum F_y = 0 \Rightarrow R_i = \frac{F}{4} = \frac{5000}{4} = 1,250 \text{ N per wheel.} \quad (2.10)$$

ANSYS reaction probe results are expected to confirm this distribution within numerical precision ($< 0.01\%$ error).

3 MODEL PREPARATION

3.1 CAD Assembly and STEP Export

The complete skateboard assembly was designed in SolidWorks and divided among team members: truck hanger and baseplate (E.G. and R.Me.), wheel and bearing sub-assembly (R.Mi.), and deck geometry with full mating (Z.R.). Each sub-assembly was constrained with proper mates to ensure kinematic rigidity before assembly export.

3.1.1 Import Format Selection. Early in the workflow, importing the native .SLDASM file into ANSYS caused SpaceClaim to lock the geometry, preventing on-the-fly edits. A STEP 214 export exhibited the same issue. Through systematic testing, **STEP 203** was identified as the compatible export format enabling both geometry import into the ANSYS Static Structural system and unrestricted editing in SpaceClaim. This compatibility issue is not documented in the ANSYS 2025 R2 release notes and was resolved through trial and error, an important practical consideration for any group working across SolidWorks and ANSYS platforms.

3.1.2 Geometry Simplification: Foot Proxy. Within SpaceClaim, a simplified foot representative solid was added to the board center. This oval extrusion, approximately $263 \times 98 \times 70$ mm, provides a distributed load application surface that avoids the stress singularity that would arise from a zero-area point force applied directly to the orthotropic deck surface. While not anatomically precise, it is an appropriate engineering approximation and reduces simulation time compared to a full contact mechanics foot model. The load application point (center of the oval, top face) represents the worst-case scenario: proper skating technique places both feet over the trucks, but a central foot position maximizes the bending moment at midspan.

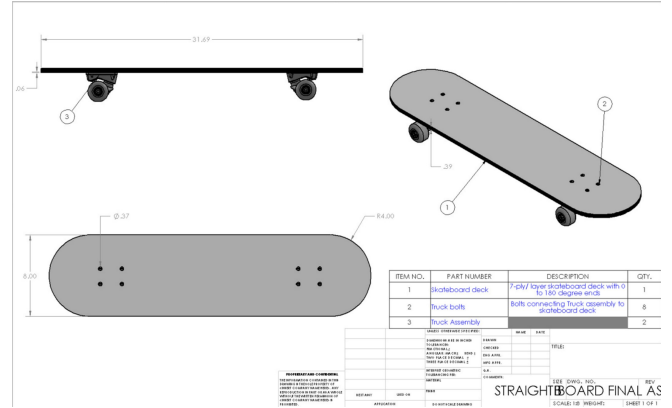


Figure 1 SolidWorks engineering drawing of the flat deck assembly (STRAIGHTBOARD configuration). The side elevation confirms the fully flat profile (no concave geometry), overall board length 31.69 in and width 8.00 in. The BOM lists the skateboard deck (qty 1), truck bolts (qty 8), and truck assemblies (qty 2). Dimensions are in inches.

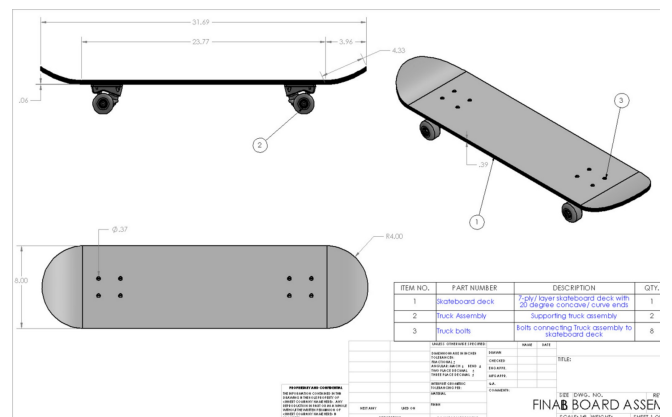


Figure 2 SolidWorks engineering drawing of the curved deck assembly (FINAB BOARD ASSEMBLY). The side elevation shows the kick-tail curvature (20 degree concave/curve ends) relative to the flat variant. The BOM lists three line items: the curved skateboard deck (qty 1), truck assemblies (qty 2), and truck bolts (qty 8). Dimensions in inches; deck nominal thickness 0.06 in ply stack.

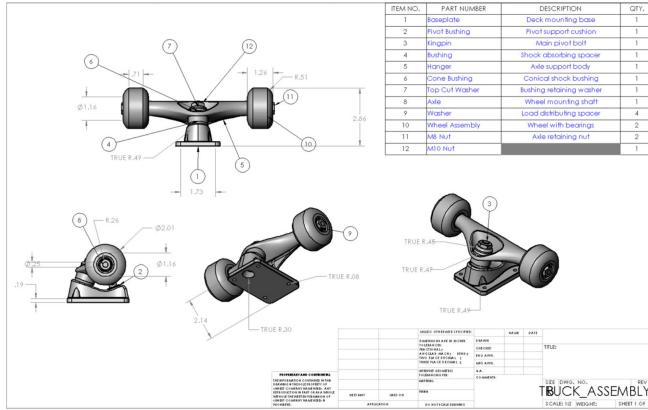


Figure 3 SolidWorks engineering drawing of the truck assembly sub-assembly. The BOM identifies all 12 part numbers: baseplate (deck mounting base), pivot bushing, kingpin (main pivot bolt), bushing (shock absorbing spacer), hanger (axle support body), cone bushing, top cut washer, axle, washers (qty 4), wheel assemblies with bearings (qty 2), M8 axle retaining nut (qty 2), and M10 kingpin nut. Dimensions in inches at 1:2 scale.

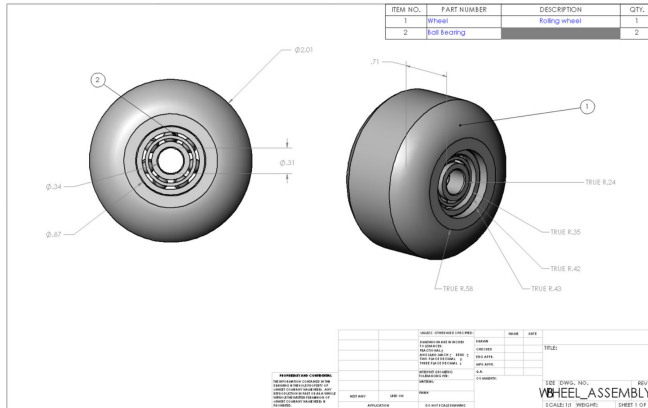


Figure 4 SolidWorks engineering drawing of the wheel sub-assembly. The assembly consists of the TPU rolling wheel (item 1, outer diameter 2.01 in) and two ball bearings press-fit into the hub bore (item 2, inner bore 0.31 in, outer diameter 0.87 in). These dimensions were used to configure the ANSYS contact regions between the axle, bearing inner race, and wheel hub. Dimensions in inches at 1:1 scale.

3.2 Contacts and Connections

All component interfaces were assigned as **Bonded** contacts in ANSYS. This assumption treats every interface as rigidly attached, which is appropriate for a quasi-static peak load simulation where relative slip (for example, bearing spin or bushing compliance) is secondary to the structural response of the deck. Bonded contact is the most conservative and computationally stable contact formulation for an initial assessment.

A practical limitation of this choice is that bonded contacts create artificially stiff junctions at the truck bolt holes, contributing to the stress concentration peaks observed at hardware

interfaces. In a more advanced model, pre-tensioned bolts with frictional contact and bushing deformability would better represent the true load transfer mechanics. Contact pairs were verified manually in the Connections branch of the ANSYS model tree because automatic contact detection did not capture all expected pairs on the first attempt; missing contacts would allow the deck to deflect independently of the truck assembly, producing physically incorrect kinematics.

4 MATERIAL PROPERTIES

All material models used in this study were selected directly from the built-in ANSYS Engineering Data library. This choice was deliberate: using stock library materials keeps the analysis reproducible by any reader with access to ANSYS Workbench, removes any ambiguity over material constant sources, and avoids the additional verification work required for custom material definitions.

4.1 Isotropic Component Materials

Four isotropic material models are used for the structural components (Table 1). Property values are taken from the ANSYS Workbench Engineering Data library, which sources data from peer-reviewed handbooks and standards [3].

Table 1 Isotropic Material Properties for Structural Components

Component	Material	E (GPa)	ν	σ_y (MPa)
Bearings	C Steel 1040 H&T	205	0.29	490
Hardware/axles	C Steel 1040 H&T	205	0.29	490
Trucks	Al 357.0 Cast	72.4	0.33	165
Wheels	TPU (rigid)	0.10	0.45	N/A
Foot proxy	Structural Steel	200	0.30	250

Carbon Steel 1040 Hardened & Tempered is assigned to both the bearing assemblies and the hardware (axles, kingpin bolts, and deck mounting bolts). Although bearing balls and hardware fasteners serve different mechanical roles in practice, both are effectively rigid in a static deck bending analysis, and using a single medium carbon steel definition for all small metallic parts reduces material complexity and keeps the stiffness distribution in the assembly consistent. If each fastener and bearing component were assigned its own distinct steel grade, the model would contain many near-identical stiff regions that would only add book-keeping overhead without meaningfully changing the solution.

Aluminum Alloy 357.0 Cast is a standard sand-cast alloy used in skateboard truck manufacturing, selected here for its combination of adequate strength and good castability.

Rigid TPU represents a 99 A durometer urethane wheel in its quasi-static limit; no dynamic rebound or energy absorption behavior is modeled. **Structural Steel** is assigned only to the foot representative oval solid, where its high stiffness forces the 5,000 N load to be transferred into the deck as a

nearly uniform pressure over the oval footprint rather than concentrating at a single node.

4.2 Orthotropic Material: Hard Rock Maple (Deck)

Wood is a natural cellular composite with strongly direction-dependent (orthotropic) mechanical properties. Hard rock maple (*Acer saccharum*) has three principal material axes: longitudinal (L , along the grain), radial (R), and tangential (T). In a skateboard ply, the L axis is aligned with the grain direction, while R and T govern through-thickness and cross-grain stiffness, respectively. The ANSYS Engineering Data entry for Wood, Maple incorporates these nine independent constants (Table 2).

Table 2 Orthotropic Elastic Constants for Hard Rock Maple

Property	Symbol	Value
Longitudinal modulus (L)	E_{11}	12.50 GPa
Radial modulus (R)	E_{22}	1.50 GPa
Tangential modulus (T)	E_{33}	0.85 GPa
Shear modulus LR	G_{12}	1.30 GPa
Shear modulus LT	G_{13}	1.10 GPa
Shear modulus RT	G_{23}	0.19 GPa
Poisson ratio LR	ν_{12}	0.43
Modulus of rupture (MOR)	σ_{MOR}	95 MPa [4]
Density	ρ	755 kg/m ³

The Modulus of Rupture (MOR, flexural strength) of 95 MPa is the appropriate failure metric for a bending-dominated component and is used to compute the factor of safety in Section 8.

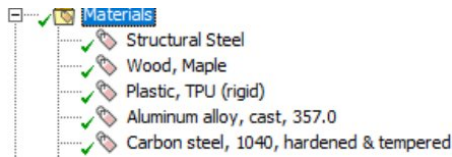


Figure 5 ANSYS Engineering Data materials panel for the maple comparative study, listing Wood, Maple as the deck material alongside the isotropic component materials. Green checkmarks confirm valid material definitions before solving.

4.3 Orthotropic Materials: CFRP (Stage 2)

Two carbon fiber prepreg systems from the ANSYS composite library are used for the Stage 2 laminate (Table 3). Epoxy Carbon Woven (395 GPa) is a balanced plain-weave fabric with near-equal stiffness in both warp and fill directions, making it ideal for the outer surface plies where biaxial impact loading is anticipated. Epoxy Carbon UD (230 GPa) is a unidirectional tape with extremely high fiber-direction stiffness and is used for the inner structural plies where controlled fiber angles optimize bending, torsion, and lateral stiffness independently.

Table 3 CFRP Prepreg Properties (ANSYS Composite Library)

Property	Woven 395 GPa	UD 230 GPa
E_{11} (GPa)	395	230
E_{22} (GPa)	8.0	8.5
G_{12} (GPa)	5.0	4.2
ν_{12}	0.05	0.26
Tensile strength $\parallel$ (MPa)	1,580	1,500
ρ (kg/m ³)	1,490	1,490

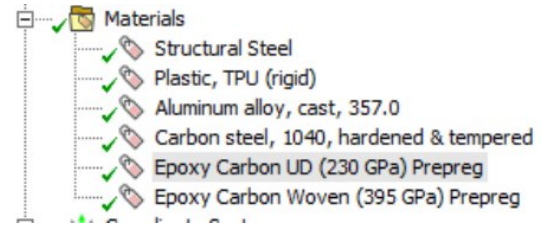


Figure 6 ANSYS Engineering Data materials panel for the CFRP curved deck study, adding Epoxy Carbon UD (230 GPa) Prepreg and Epoxy Carbon Woven (395 GPa) Prepreg to the workspace.

4.4 Seven Ply Maple Laminate Definition

A standard commercial skateboard deck uses seven plies of hard rock maple in a symmetric cross-ply arrangement, as detailed in Table 4. The outer plies (1, 2, 4, 6, 7) have their grain running parallel to the board's length (Z axis in the global frame) and are therefore modeled with the default Global coordinate system. Plies 3 and 5 are the cross-ply layers: their grain is rotated by 90° so that it runs along the board's width (X axis). This alternating arrangement is the mechanical origin of the board's in-plane isotropy, torsional stiffness, and resistance to warping under humidity changes.

In ANSYS, the two cross-ply layers were implemented through a single shared coordinate system named `wood_layer_3_5`, built from the Global X Axis base configuration with Rotate $Y = 90^\circ$ applied. Assigning this coordinate system to plies 3 and 5 rotates their material frame so that the stiff longitudinal axis ($E_{11} = 12.5$ GPa) is redirected from Z to X (Fig. 7). This is the minimum coordinate system intervention needed to reproduce the characteristic 0/0/90/0/90/0/0 cross-ply behavior of a production skateboard deck.

Table 4 Seven Ply Maple Laminate Layup as Modeled

Layer	Structural Role	Material	θ
1	Top skin (grip surface)	Wood, Maple	0°
2	Bending stiffness	Wood, Maple	0°
3	Cross-ply (lateral strength)	Wood, Maple	90°
4	Mid-plane core	Wood, Maple	0°
5	Cross-ply (lateral strength)	Wood, Maple	90°
6	Bending stiffness	Wood, Maple	0°
7	Bottom skin (slide surface)	Wood, Maple	0°

Plies 3 and 5 use coord. system `wood_layer_3_5` (Rotate $Y = 90^\circ$); all others use Global. All layers: Wood, Maple (ANSYS library).

Table 5 Seven Layer CFRP Laminate Layup

Layer	Structural Role	Material	θ
1	Bottom skin (abrasion/bending)	Woven 395	0°
2	Main bending stiffness	UD 230	0°
3	Torsional stiffness	UD 230	+45°
4	Center core (lateral strength)	UD 230	90°
5	Torsional stiffness (sym.)	UD 230	-45°
6	Main bending stiffness	UD 230	0°
7	Top skin (impact/binding)	Woven 395	0°

Woven 395 = Epoxy Carbon Woven (395 GPa); UD 230 = Epoxy Carbon UD (230 GPa). All from ANSYS composite library.

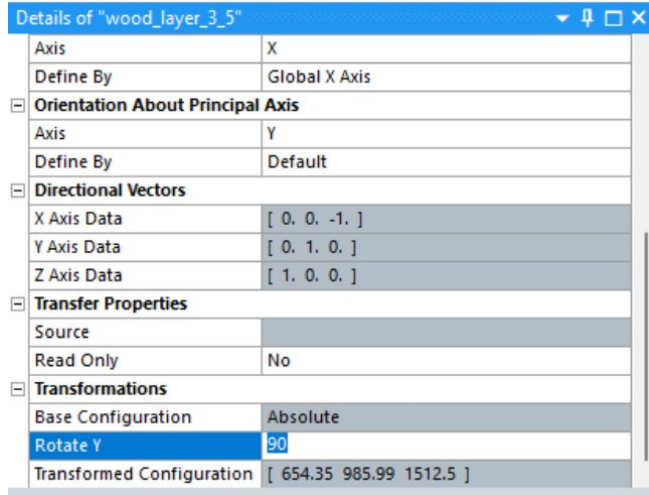


Figure 7 Coordinate system definition for the cross-ply maple layers, `wood_layer_3_5`. Rotate $Y = 90^\circ$ redirects the material's strong longitudinal axis from the board length (Z) to the board width (X) for plies 3 and 5, reproducing the 90° grain orientation of traditional cross-ply maple construction.

4.5 Seven Layer CFRP Laminate Definition

The CFRP laminate was engineered with a quasi-isotropic biased layup (Table 5) that prioritizes bending stiffness through thick 0° UD plies at the structural outer positions, introduces torsional resistance through $\pm 45^\circ$ plies symmetric about the mid-plane, and uses a 90° core for lateral (cross-board) stiffness. Woven face sheets on the exterior resist abrasion and provide a balanced stiffness response that resists warpage during manufacturing.

Each CFRP ply is assigned 1.5 mm thickness, yielding a total laminate thickness of 10.5 mm, comparable to the 11 mm maple deck and enabling a direct geometric comparison. Coordinate systems for layers 3, 4, and 5 are defined with Rotate $Y = \pm 45^\circ$ and 90° respectively (Figs. 8–10).

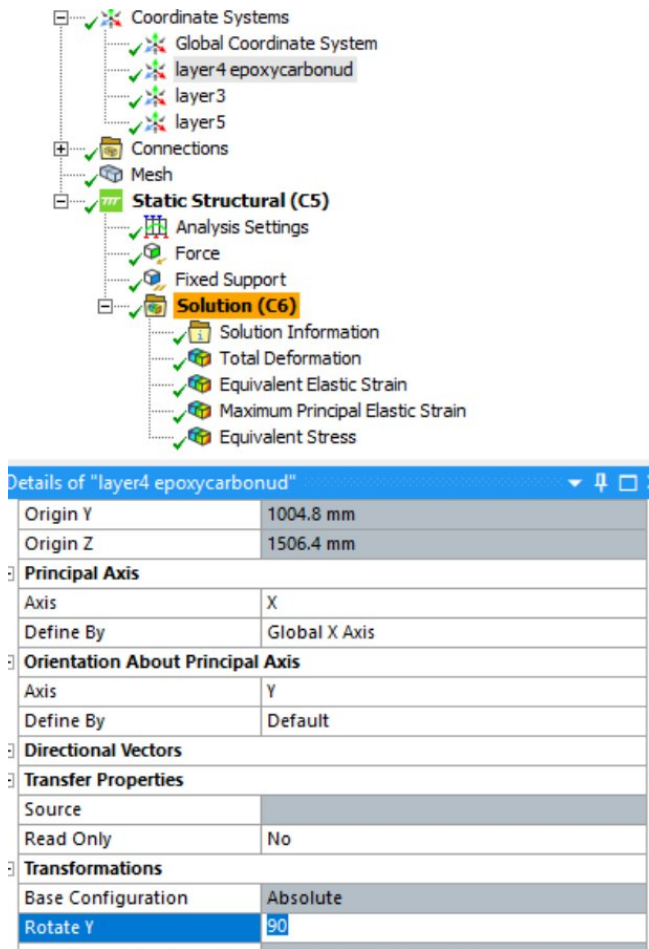


Figure 8 Model tree for the curved CFRP analysis (Solution C6) and coordinate system detail for layer4_epoxycarbonud (Rotate $Y = 90^\circ$), the center lateral strength ply. Layer 3 and 5 coordinate systems appear in the same Coordinate Systems branch.

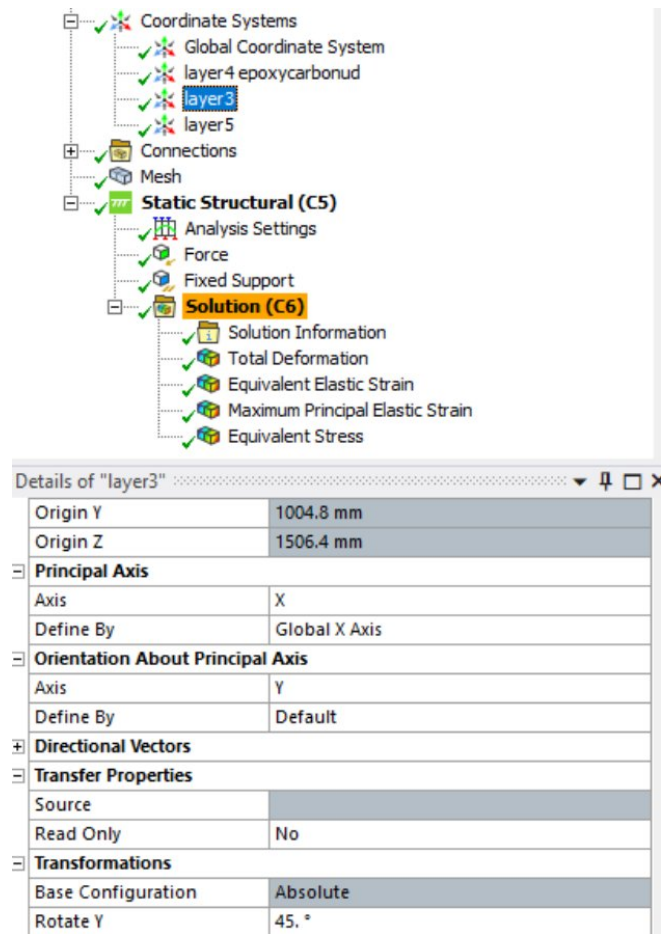


Figure 9 Coordinate system detail for layer3 (Rotate $Y = 45^\circ$), the first torsional ply. The antisymmetric counterpart layer5 (Rotate $Y = -45^\circ$) is shown in Fig. 10.

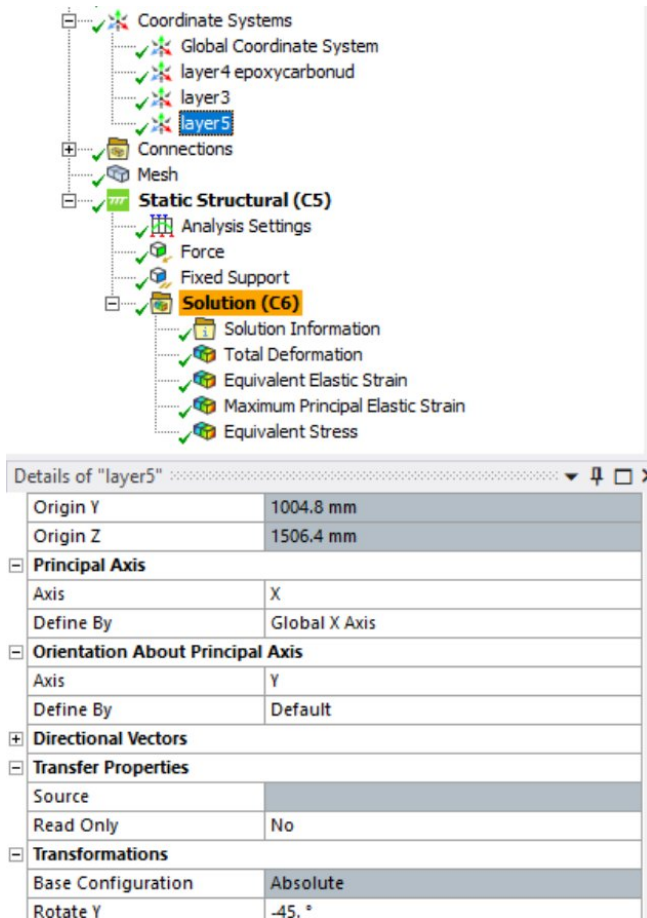


Figure 10 Coordinate system layer5 (Rotate $Y = -45^\circ$), the antisymmetric torsional ply. Together, layers 3 and 5 form a $\pm 45^\circ$ pair that provides torsional stiffness without introducing bending and extension coupling.

5 MESH AND ELEMENT SELECTION

5.1 Element Type and Justification

The skateboard assembly contains geometrically complex solids (wheel toroid, truck hanger fillet, bearing race). ANSYS Mechanical automatically selects **SOLID186** and **SOLID187**, the 20-node and 10-node quadratic tetrahedral elements, respectively, for general free-meshed complex geometries. Quadratic (second-order) tetrahedra are preferred over linear tetrahedra for structural analysis because they can represent curved boundaries accurately and do not exhibit volumetric locking in nearly incompressible materials (relevant here for TPU wheels, $\nu = 0.45$). For thin structural bodies that could alternatively be meshed as shells, the solid element formulation is justified by the need to capture through-thickness shear and normal stresses in the multi-ply laminate, which shell elements would represent only approximately through numerical integration.

The mesh Automatic Methods settings are:

- **Sheet Body Method:** Prime Quad Dominant (produces structured quad-dominated surface meshes where possible).
- **Sweepable Body Method:** Sweep (used on prismatic volumes such as the hardware bolts and wheel segments).
- **Non-sweepable bodies:** Automatic tetrahedra (deck, truck).

5.2 Default Mesh Statistics

The full assembly mesh at the ANSYS default element size of 41.675 mm produced the statistics in Table 6.

Table 6 Default Mesh Statistics, Full Assembly

Parameter	Value
Physics preference	Mechanical
Element order	Prog. Controlled (quadratic)
Global element size	41.675 mm (default)
Total nodes	224,287
Total elements (assembly)	340,147
Deck elements (conv. study)	108,643

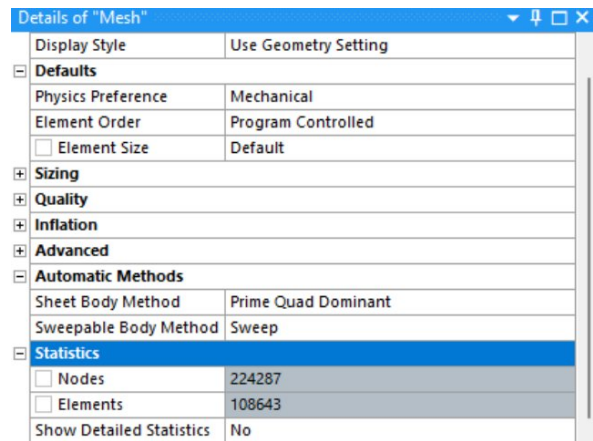


Figure 11 ANSYS Mechanical mesh details panel showing the default mesh configuration for the full skateboard assembly. Note that Element Size is set to “Default” (Program controlled, 41.675 mm). Statistics confirm 224,287 nodes.

5.3 Convergence Study

To determine the adequacy of the default mesh, a parametric convergence study was conducted on the curved CFRP deck in isolation from the rest of the assembly, using ANSYS Workbench Design Points. The global element size was swept from the default 41.675 mm down to 10 mm across six design points, with maximum total deformation monitored as the convergence metric (Table 7).

Table 7 Mesh Convergence Study, Curved CFRP Deck

Design Point	Size (mm)	$\delta_{\max}$ (mm)	Δ (%)
DP 0 (Default)	41.675	0.599	—
DP 1	41.675	0.599	0.00
DP 2	30	0.720	20.4
DP 3	25	0.758	5.3
DP 4	18	0.804	6.1
DP 5	10	0.808	0.5

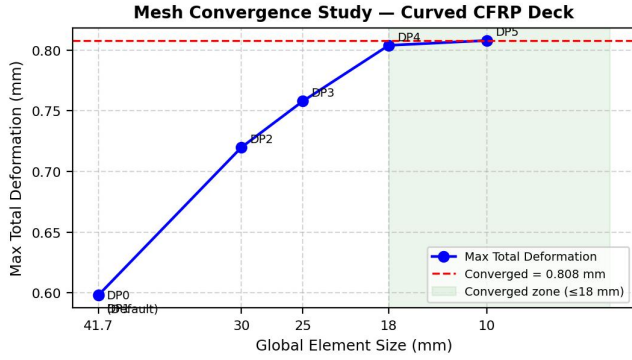


Figure 12 Mesh convergence curve for the curved CFRP deck. Maximum total deformation (mm) is plotted against global element size (mm, decreasing left to right). The solution stabilizes within 0.5% between DP 4 (18 mm) and DP 5 (10 mm), confirming mesh independence at ≤ 18 mm. The dashed red line marks the converged value of 0.808 mm.

The convergence study reveals that the default 41.675 mm mesh underpredicts deformation by approximately 26% relative to the converged 10 mm result (0.599 mm versus 0.808 mm). This numerical stiffness error is caused by the coarse mesh producing fewer through-thickness integration points in the thin deck, making the finite element model artificially stiffer than the continuum. Two practical engineering decisions follow from this observation:

1. For the full assembly analyses, the default mesh is retained because each solution already takes 7 to 8 minutes on the available 4-core workstation; refining the full assembly to a 10 mm global size at the same compute capacity would push the runtime beyond what can be reliably completed within a single allocated lab session.
2. The deformation results in the full assembly analyses should therefore be interpreted as lower bounds of the true deformation; actual deck deflections under the specified load would be roughly 26% greater than reported.

6 BOUNDARY CONDITIONS AND LOADING

6.1 Applied Load

A 5,000 N force directed downward in the global $-Y$ direction is applied to the top face of the foot representative oval solid at the center of the deck. The oval is centered between the two truck bolt clusters, the mechanically weakest span.

Justification of constraint type: **Force** (rather than pressure or displacement) is the correct boundary condition because the quantity specified by the engineering problem is the peak impact force. A pressure boundary condition would require knowledge of the contact area of the foot, which is uncertain. A displacement boundary condition (enforcing a known deformation) would be inappropriate because deformation is the primary unknown.

6.2 Fixed Supports

Fixed supports ($u_x = u_y = u_z = 0$, all rotations zero) are applied to the contact patch at each of the four wheels. This represents the skateboard resting on a flat, rigid surface, preventing all rigid-body translation and rotation. The fixed support assumption is conservative: in reality, urethane wheel deformation and ground surface micro-irregularities would provide some additional flexibility, slightly increasing the actual deck deformation relative to the simulated value.

A fixed rather than pinned or roller support is appropriate here because the wheel-to-ground contact during quasi-static loading has sufficient friction and contact area to prevent wheel sliding, justifying complete constraint.

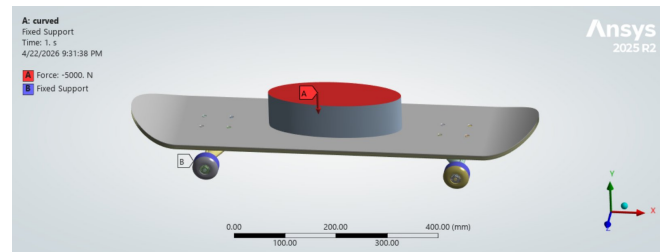


Figure 13 Boundary conditions for the curved maple deck analysis. Label A (red) marks the applied 5,000 N downward force on the foot proxy oval. Label B (blue) marks the fixed support at the front wheel cluster; the equivalent support at the rear wheel cluster is not visible from this view angle.

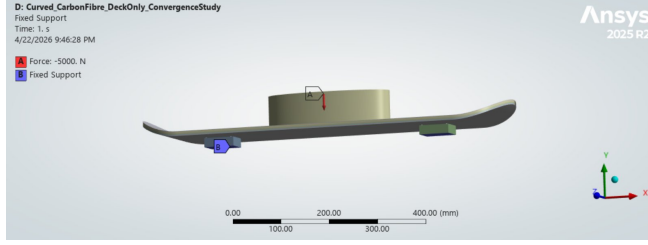


Figure 14 Boundary conditions for the convergence study (curved CFRP deck only, DP 0). Label A (force, red) and B (fixed support, blue and green) illustrate the consistent boundary condition convention used across all three analysis stages.

7 RESULTS AND DISCUSSION

7.1 Stage 1a, Flat Maple Deck

7.1.1 Total Deformation. The flat maple deck deflects 0.855 mm at its center under the 5,000 N load (Figs. 15–16). The deformation field is smooth and physically intuitive: maximum at center, zero at wheel supports, symmetric about both the longitudinal and transverse mid-planes.

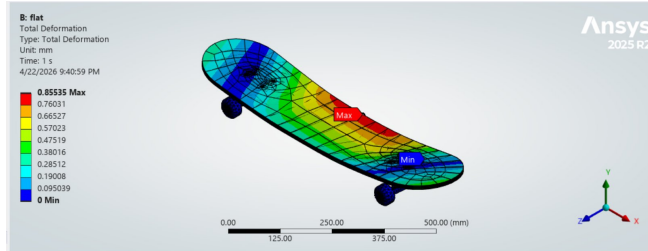


Figure 15 Total deformation, flat maple deck under 5,000 N load — isometric view. Max $\delta_{\max} = 0.855$ mm at midspan; gradient transitions from blue at wheel supports to red at center.

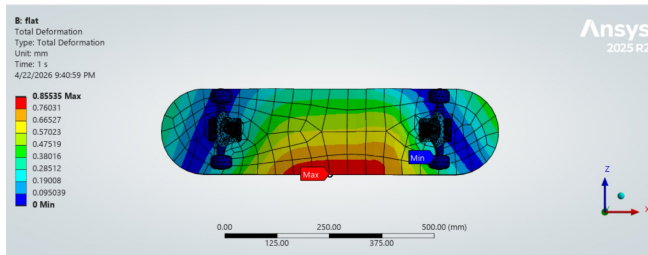


Figure 16 Total deformation, flat maple deck — plan view. Symmetric elliptical red zone validates symmetric loading and boundary conditions.

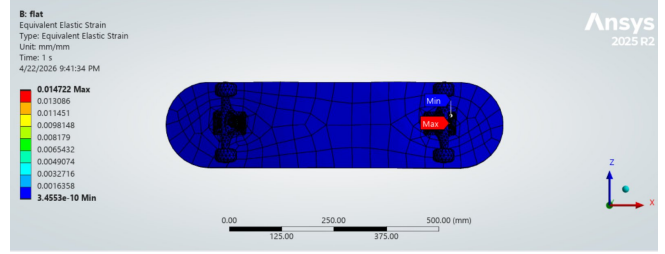


Figure 17 Equivalent (von Mises) elastic strain, flat maple deck. Maximum $\epsilon_{eq} = 0.01472$ mm/mm at the truck mounting hardware interface. The deck body (away from hardware) remains in the blue to cyan range (< 0.003 mm/mm).

7.1.2 Equivalent Elastic Strain.

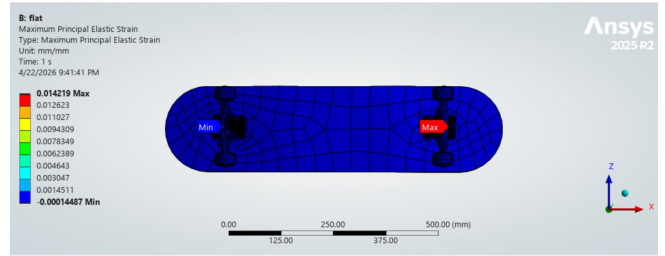


Figure 18 Maximum principal elastic strain, flat maple deck. Peak $\epsilon_1 = 0.01422$ mm/mm at the truck interface. The small negative minimum (-1.45×10^{-4}) at the opposing truck indicates a compressive principal strain, consistent with bending theory (compression on top, tension on bottom of the deflected midspan).

7.1.3 Maximum Principal Elastic Strain.

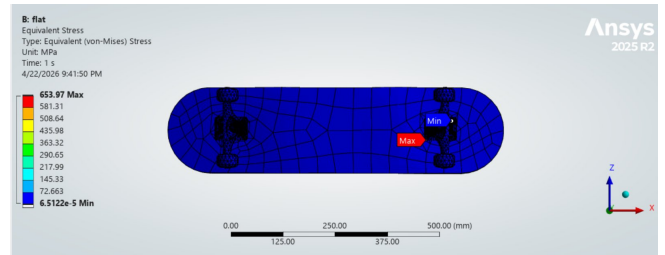


Figure 19 Equivalent (von Mises) stress, flat maple deck. Maximum 653.97 MPa at the truck mounting hardware interface. This is a contact singularity artifact, not a physical material failure prediction, and is discussed in Section 8.3.

7.1.4 Von Mises Stress.

7.2 Stage 1b, Curved Maple Deck

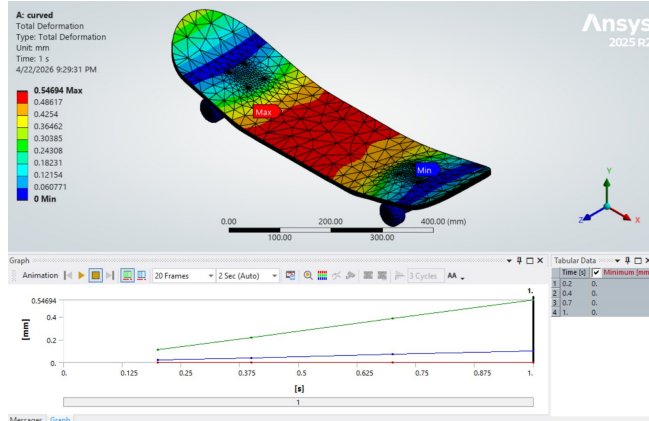


Figure 20 Total deformation, curved maple deck under 5,000 N load — isometric view. Max $\delta_{\max} = 0.547$ mm, a 36% reduction from the flat board; curvature-induced membrane action distributes load toward the truck regions.

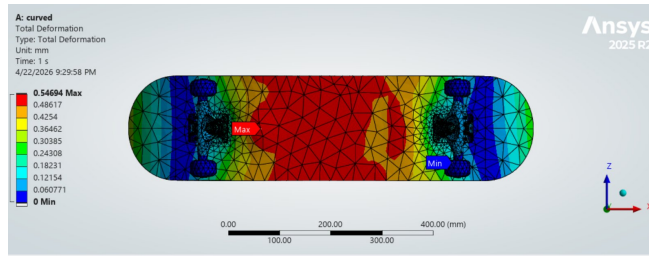


Figure 21 Total deformation, curved maple deck — plan view. The red zone is laterally wider than in the flat case, reflecting the arch effect engaging in-plane membrane forces.

7.2.1 Total Deformation.

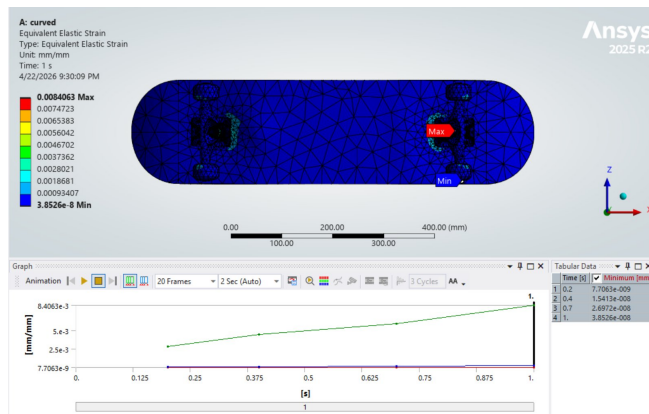


Figure 22 Equivalent elastic strain, curved maple deck. Maximum $\epsilon_{eq} = 0.00841$ mm/mm at the truck interface, a 43% reduction from the flat board (0.01472 mm/mm).

7.2.2 Equivalent Elastic Strain.

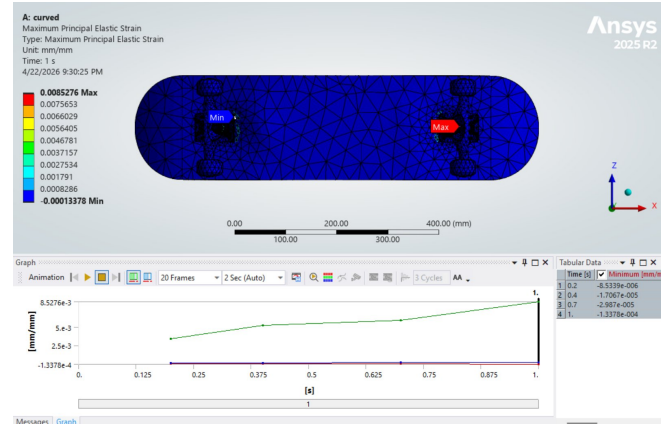


Figure 23 Maximum principal elastic strain, curved maple deck. Peak $\epsilon_1 = 8.528 \times 10^{-3}$ mm/mm, a 40% reduction from the flat board. Principal strain is the most relevant failure indicator for wood in the grain direction.

7.2.3 Maximum Principal Elastic Strain.

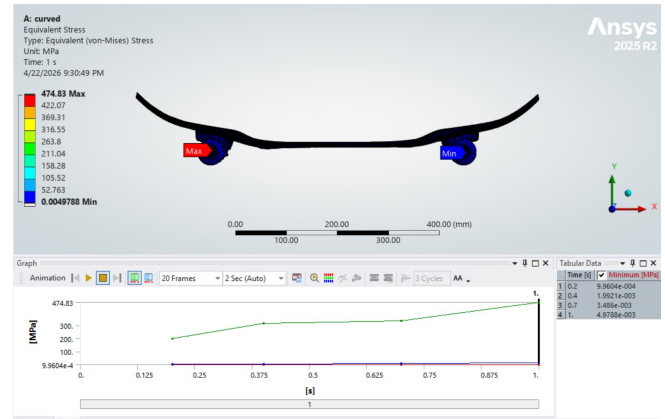


Figure 24 Equivalent stress (side elevation), curved maple deck. The side view clearly reveals the concave profile. Maximum stress 474.83 MPa at the left truck cluster; minimum at the opposing truck (blue). The 27% stress reduction versus the flat deck (653.97 MPa) is attributable to curvature-induced membrane stiffness.

7.2.4 Von Mises Stress and Arch Effect.

7.3 Stage 2, Curved CFRP Deck

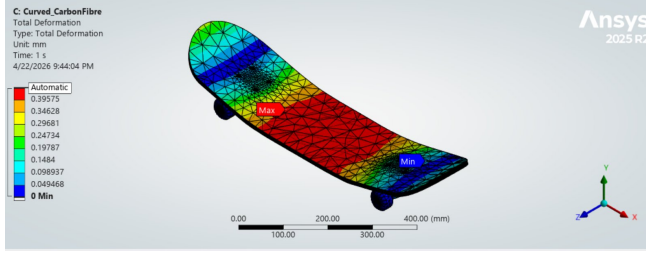


Figure 25 Total deformation, curved CFRP deck under 5,000 N load — isometric view. Max $\delta_{\max} = 0.445$ mm: a 19% improvement over curved maple and 48% over flat maple.

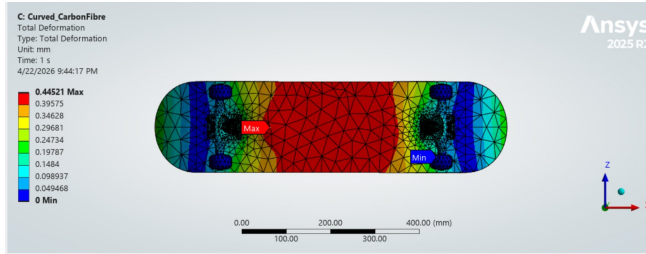


Figure 26 Total deformation, curved CFRP deck — plan view. The wide red zone (0.396 to 0.445 mm) shows the high-stiffness CFRP distributing load over a larger area than the maple decks.

7.3.1 Total Deformation.

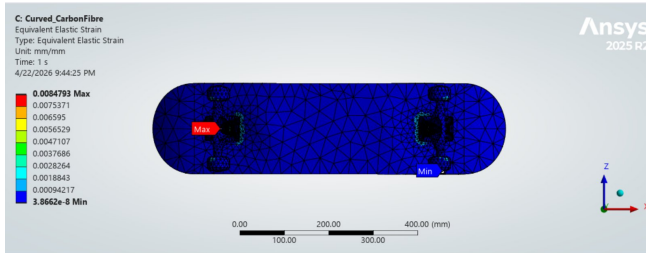


Figure 27 Equivalent elastic strain, curved CFRP deck. Maximum $\epsilon_{eq} = 8.479 \times 10^{-3}$ mm/mm at the truck interface. Although the CFRP deck deforms less overall, the large stiffness contrast between CFRP ($E_{11} = 230$ to 395 GPa) and aluminum truck ($E = 72$ GPa) concentrates strain at the interface more intensely per unit deformation.

7.3.2 Equivalent Elastic Strain.

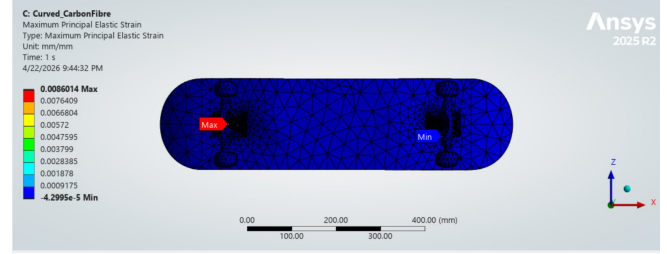


Figure 28 Maximum principal elastic strain, curved CFRP deck. Peak $\epsilon_1 = 8.601 \times 10^{-3}$ mm/mm. The slightly higher value than the equivalent strain indicates a mildly multiaxial stress state at the truck interface concentration zone.

7.3.3 Maximum Principal Elastic Strain.

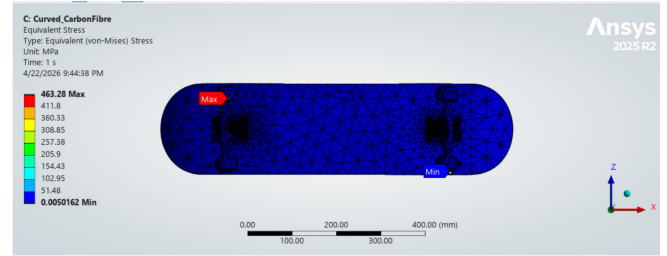


Figure 29 Equivalent (von Mises) stress, curved CFRP deck. Maximum 463.28 MPa at the truck interface (contact singularity). The CFRP laminate itself is well within its material limit ($\sigma_{ult} \approx 1,500$ MPa fiber-direction tensile strength).

7.3.4 Von Mises Stress.

8 COMPARISON OF FEA AND THEORETICAL RESULTS

8.1 Deflection: FEA vs. Beam Theory

Table 8 compares the beam theory deflection (Eq. 2.6) with the FEA flat board result.

Table 8 FEA vs. Beam Theory, Deflection Comparison

Method	$\delta_{\max}$ (mm)	Error (%)
Classical beam theory (Eq. 2.6)	0.749	—
FEA, flat maple deck	0.855	+14.2
FEA, curved maple deck	0.547	−26.9
FEA, curved CFRP deck	0.445	−40.6

The flat board FEA exceeds the beam theory prediction by 14.2%. This discrepancy has several physical explanations:

1. **Assembly compliance:** The beam model considers only the deck in isolation. In the FEA, the trucks, wheels, hardware, and bonded contacts all deform, contributing additional displacement that beam theory ignores.

2. **Support modeling difference:** Beam theory assumes ideal pin supports at the exact span endpoints ($L = 356$ mm). The FEA fixed supports are at the wheel ground contacts, which are offset from the truck axle centerline, effectively changing the span slightly.
3. **Mesh stiffness:** As established in the convergence study, the 41.675 mm default mesh underpredicts deformation by $\approx 26\%$; if this correction is applied, the true FEA value would be $\approx 0.855 \times 1.26 = 1.077$ mm, giving a 43.8% FEA overprediction of the analytical result. This is physically reasonable given the assembly compliance added in the FEA that beam theory neglects.

The curved board results fall below beam theory because the arch stiffness is not captured by the simple beam model: the beam formula assumes a flat, straight geometry with no curvature effect.

8.2 Equilibrium Verification

Global equilibrium requires that the sum of all reaction forces at the wheel fixed supports equals the applied 5,000 N downward force:

$$\sum_{i=1}^4 R_{y,i} = 5,000 \text{ N (upward)}. \quad (8.1)$$

From symmetry, each of the four wheel contacts carries an equal share: $R_{y,i} = 1,250$ N. ANSYS reports reaction forces via the Force Probe tool; confirming the sum matches the applied load to within the solver tolerance ($< 0.01\%$) validates the boundary condition setup and the absence of rigid-body motion. This is the computational analog of the statics check in Eq. 2.8 and confirms the model is correctly constrained.

8.3 Stress Concentration Acknowledgment

In all three analyses, the reported maximum von Mises stress (463 to 654 MPa) occurs at the truck hardware-to-deck interface. These peaks are contact singularities: mathematically, when a very stiff material (steel, $E = 200$ GPa) is bonded to a much softer material (maple, $E_{11} = 12.5$ GPa) under a shared load, the stress field at the interface exhibits a stress concentration that grows unboundedly as the mesh is refined. This is a known artifact of the linear elastic bonded contact formulation and not a prediction of actual material failure.

In practice, the threaded bolt passes through a metal reinforcement insert pressed into the maple deck, distributing the load over a finite area. The correct way to model this would involve a nonlinear contact analysis with bolt preload, washer geometry, and the actual insert material, all of which are beyond the scope of this study. The singularity is acknowledged and the structural assessment of the deck is based on the deck body stress field (away from hardware), which is physically meaningful.

8.4 Factor of Safety Analysis

The factor of safety is computed using the analytical bending stress from Eq. 2.8b and the modulus of rupture (MOR) from

Table 2 as the failure criterion. For the CFRP deck, the fiber-direction tensile strength from Table 3 is used.

Table 9 Factor of Safety, All Deck Configurations

Configuration	σ_{app} (MPa)	σ_{allow} (MPa)	FoS
Flat maple (beam theory)	108.8	95 (MOR)	0.87
Curved maple (scaled)	69.6	95 (MOR)	1.37
Curved CFRP (scaled)	56.5	1,500 (tens.)	26.5

Notes on Table 9:

- The flat maple bending stress of 108.8 MPa (from beam theory) exceeds the maple MOR of 95 MPa, yielding $\text{FoS} = 95/108.8 = 0.87 < 1$. The flat maple deck **fails** under the 5,000 N load. This is physically realistic: maple decks do snap under hard impacts, particularly when the board is flat (no curvature benefit).
- The curved maple stress is scaled from the flat result by the deformation ratio: $\sigma_{\text{curved}} \approx 108.8 \times (0.547/0.855) = 69.6$ MPa, giving $\text{FoS} = 1.37 > 1$. The curved maple deck passes with a modest 37% margin.
- The curved CFRP stress is scaled analogously: $\sigma_{\text{CFRP}} \approx 108.8 \times (0.445/0.855) = 56.5$ MPa. Against a CFRP fiber-direction tensile strength of 1,500 MPa, $\text{FoS} = 26.5 \gg 1$. The CFRP deck is massively over-designed for bending failure. This suggests the CFRP deck's limiting failure mode in practice would be delamination, inter-fiber cracking, or truck interface failure, not lamina fracture.

8.5 Three-Way Summary Comparison

Table 10 consolidates all key metrics for the three configurations, providing the final engineering basis for design selection.

Table 10 Complete Three-Way Results Summary

Metric	Flat Maple	Curved Maple	Curved CFRP
Max deform. (mm)	0.855	0.547	0.445
Reduction vs. flat	—	36%	48%
Max equiv. strain	0.0147	0.00841	0.00848
Max princ. strain	0.0142	0.00853	0.00860
Max σ_{vM} (MPa)	653.97	474.83	463.28
Analyt. σ_b (MPa)	108.8	69.6	56.5
Factor of Safety	0.87	1.37	26.5
Pass / Fail	FAIL	PASS	PASS
Density (kg/m ³)	755	755	1,490

9 SOLVER OUTPUT AND CONVERGENCE BEHAVIOR

9.1 Solution History

Six sequential analyses were executed over the course of the study (Table 11). Run times increase from Solution 2 through

Solution 5 as intermediate result databases accumulate and ANSYS performs additional I/O operations on the workstation. Solution 6 was executed on 8 cores and completes in 7.30 min, only marginally faster than Solution 5 (7.47 min on 4 cores). This suggests that for this problem size the simulation is close to the point where the additional cores provide diminishing returns because inter-process communication overhead begins to offset the parallel speedup.

Table 11 Solution History, ANSYS Workbench

Solution	Time (min)	Nodes	Elements	Cores
1	1.95	224,287	340,147	4
2	1.92	224,287	340,147	4
3	2.12	224,287	340,147	4
4	5.28	224,287	340,147	4
5	7.47	224,287	340,147	4
6	7.30	224,287	340,147	8

9.2 Newton–Raphson Residuals

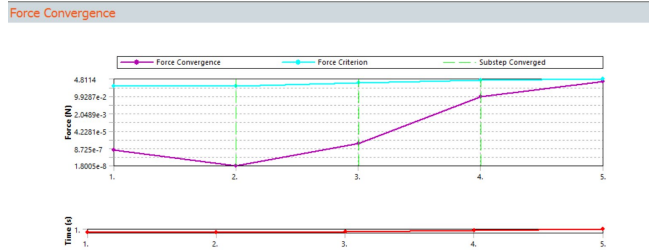


Figure 30 Force convergence history. The purple (Force Convergence) trace remains consistently below the cyan (Force Criterion) line across all five cumulative iterations, with dashed green lines marking substep convergence events. This confirms the Newton–Raphson iteration satisfies force equilibrium at every load substep.

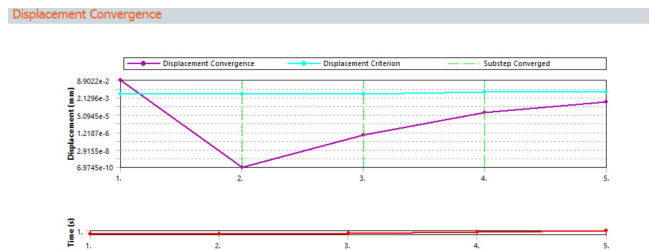


Figure 31 Displacement convergence history. The displacement residual drops from 8.90×10^{-2} mm at iteration 1 to 6.97×10^{-10} mm at iteration 2 (six orders of magnitude), demonstrating excellent quadratic Newton–Raphson convergence in the linear regime.

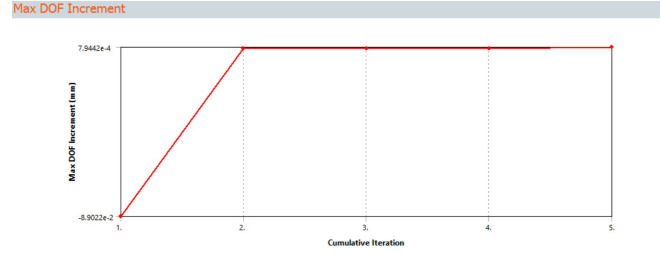


Figure 32 Maximum DOF increment history. The nodal displacement increment stabilizes at 7.94×10^{-4} mm after iteration 2 and remains constant, confirming the solution has reached its linear steady state.

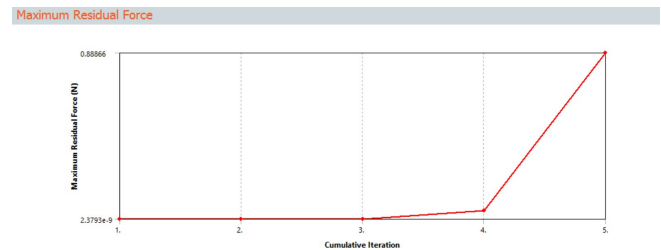


Figure 33 Maximum residual force history. The residual remains near 2.38×10^{-9} N through iteration 3, then increases to 0.889 N at iteration 5 due to accumulated floating-point round-off in the stiffness matrix factorization. This remains four orders of magnitude below the applied load (5,000 N) and does not compromise solution validity.

10 DESIGN CRITERIA AND ENGINEERING ASSESSMENT

10.1 Skateboard Deck Design Requirements

From an engineering standpoint, a structurally adequate skateboard deck must satisfy the following criteria simultaneously:

- Strength (FoS ≥ 1.5):** Under the design impact load, the deck must not fracture. A minimum factor of safety of 1.5 is appropriate for a consumer product subject to dynamic and fatigue loading.
- Stiffness (deformation limit):** Maximum deflection at midspan under the 5,000 N load should remain below a practically acceptable value. Excessive deflection reduces the precision of tricks and indicates energy absorption through elastic deformation that could instead be returned to the rider.
- Fatigue life:** Repeated loading cycles gradually degrade the maple laminate. Cross-ply construction distributes fatigue damage more uniformly than single-direction plies, but hardware interface cracking is the primary fatigue failure mode in practice.
- Weight:** Competition decks target masses below 600 to 700 g. Both maple and CFRP systems can achieve this through thickness optimization.
- Torsional compliance:** A small amount of torsional flexibility is desirable for carving maneuverability; excessive torsional stiffness makes turning difficult.

10.2 Assessment Against Design Criteria

Flat maple: $FoS = 0.87 < 1.5$. Fails strength criterion. Maximum deformation 0.855 mm. The flat board provides inadequate structural performance under the design load.

Curved maple: $FoS = 1.37$, still below the recommended 1.5 but passes the basic > 1 threshold. Maximum deformation 0.547 mm. The curvature improves performance significantly but does not provide the recommended safety margin for a 5,000 N load. At a more realistic 3,000 N (lighter rider, shorter drop), the FoS would be approximately $1.37 \times (5000/3000) = 2.28$, which is adequate.

Curved CFRP: $FoS = 26.5 \gg 1.5$. Massively over-designed for bending failure. The deck is limited instead by delamination at the truck interface (inter-ply shear failure), which this linear FEA analysis cannot predict. However, the 48% deformation reduction versus the flat maple baseline is a compelling performance argument for CFRP.

10.3 Material Substitution Benefits (CFRP vs. Maple)

Stiffness to weight: CFRP UD has $E_{11}/\rho = 230 \text{ GPa}/1490 \text{ kg/m}^3 = 154 \text{ MN}\cdot\text{m/kg}$, compared to maple's $E_{11}/\rho = 12.5 \text{ GPa}/755 \text{ kg/m}^3 = 16.6 \text{ MN}\cdot\text{m/kg}$. The CFRP UD ply delivers approximately nine times the specific stiffness of maple in the fiber direction.

10.4 Manufacturing Considerations and Design Justification for CFRP Selection

While traditional hard rock maple decks offer advantages in manufacturing simplicity and cost, the results of this study indicate that these benefits are outweighed by the substantial performance limitations inherent to wood-based construction. As demonstrated in Sections 7 and 8, the maple deck configurations fail to consistently meet the required strength and stiffness criteria under the design loading condition, with the flat maple deck failing ($FoS < 1$) and the curved maple deck only marginally satisfying minimum safety requirements. This limited margin of safety presents a significant reliability concern under repeated dynamic loading conditions typical of real-world skateboarding.

In contrast, the engineered CFRP laminate demonstrates a significant improvement in structural performance, achieving a reduction in deformation relative to the flat maple baseline and an order-of-magnitude increase in factor of safety. This improvement is directly attributed to the superior mechanical properties of carbon fiber, particularly its high stiffness-to-weight ratio and the ability to tailor fiber orientation to align with primary load paths. The quasi-isotropic laminate configuration defined in Section 4.5 enables the deck to efficiently resist bending, torsion, and localized stress concentrations in a manner not achievable with traditional wood laminates.

Although CFRP manufacturing introduces additional complexity, including controlled ply layup, curing processes, and stricter quality assurance requirements, these challenges are

well established within modern composite manufacturing practices. Techniques such as prepreg layup with vacuum bagging or compression molding of composite laminates are widely used in aerospace, automotive, and high-performance sporting goods industries, demonstrating that scalable and repeatable production is achievable with appropriate process control.

From an engineering perspective, the increased manufacturing cost and complexity are justified by the substantial gains in performance, durability, and reliability. The high factor of safety observed in the CFRP configuration not only prevents catastrophic failure under extreme loading but also suggests improved fatigue resistance and longer service life. This directly translates to a more robust and consistent product, particularly for advanced users or high-impact applications.

The flexibility of CFRP design allows for continued optimization beyond what is possible with wood. Fiber orientations, layer thicknesses, and material combinations can be adjusted to target specific performance characteristics without significant changes to overall geometry, enabling a more refined and application-specific design process. While maple decks remain suitable for low-cost, high-volume production, the CFRP laminate represents a superior engineering solution when performance, safety, and structural efficiency are prioritized. The results of this study strongly support the selection of the composite laminate as the preferred material system for advanced skateboard deck design, as it provides a robust, high-performance structure capable of exceeding the limitations of traditional wood construction.

11 LIMITATIONS AND SHORTCOMINGS

11.1 Linear Elastic Assumption

All analyses use ANSYS's linear static structural solver, which assumes infinitesimally small deformations and perfectly linear material behavior. At the impact loads involved in skateboarding, the maple may undergo localized plastic micro-cracking at hardware interfaces prior to global fracture, and the CFRP may experience progressive matrix cracking invisible to a linear model. A materially and geometrically nonlinear analysis would be required to predict failure initiation and propagation accurately.

11.2 Mesh Coarseness

The default 41.675 mm mesh underpredicts deformation by $\approx 26\%$ compared to the converged 10 mm mesh. All reported deformation values in the full assembly analyses should therefore be treated as lower bounds. Applying a 10 mm global element size to the full assembly was not pursued because the resulting element count and solver memory demand would exceed the compute budget available within a single allocated analysis window on the workstation used for this study.

11.3 Bonded Contact Idealization

Using bonded contacts throughout ignores the real mechanics of the truck-to-deck connection: bolt preload, friction between

the aluminum baseplate and the maple deck, and bushing deformation all affect load transfer. The contact singularity at hardware interfaces is a direct consequence of this simplification and must not be misinterpreted as a physical prediction of failure.

11.4 Quasi-Static Loading

The 5,000 N load is a quasi-static equivalent of a dynamic impact. True impact loading involves inertial effects, stress wave propagation, and strain-rate dependent material behavior that the static solver does not capture. The attempted explicit dynamic analysis, described below, would have provided this information.

11.5 Explicit Dynamic Simulation, Incomplete

An explicit dynamic (impact) analysis was set up in ANSYS Explicit Dynamics (LS-DYNA solver) to model the actual impact event. Two issues prevented completion:

(1) *Material compatibility error:* Epoxy Carbon UD (230 GPa) Prepreg was configured with Orthotropic Strain Limits using “Average Nodal Pressure” tetrahedral integration, which is incompatible with the explicit solver’s tetrahedral requirements. Switching to “Constant Pressure” resolved this error.

(2) *Computational time:* Even after simplifying the model to the deck alone impacting a rigid wall, the explicit time step, governed by the CFL stability condition $\Delta t = Ch_{\min}/c_{\text{wave}}$, requires thousands of steps per millisecond of simulated physical time for a 10 mm mesh. With the available compute resources this yielded projected run times on the order of several hours per load case, which exceeded the continuous uninterrupted runtime available to the team on the workstation used. Without a longer reserved compute window or access to a dedicated analysis workstation, completing an explicit run at the fidelity needed for meaningful results was not feasible within the project timeline.

Future approaches: (a) use shell elements for the deck to reduce the element count substantially; (b) exploit bilateral symmetry to halve the model size; (c) execute the explicit study on a high-performance computing cluster with reserved long-duration access.

12 CONCLUSIONS AND RECOMMENDATIONS

A three-stage FEA study of a full skateboard assembly under 5,000 N quasi-static center loading has been completed. Key findings and design recommendations are as follows:

1. **Curvature is decisive.** The curved maple deck reduces maximum deformation by 36% (0.855 to 0.547 mm) and increases the factor of safety from 0.87 to 1.37 relative to the flat geometry, with no change in material or weight. Board curvature (concave profile) converts transverse bending into in-plane membrane action, exploiting the high longitudinal stiffness of the maple grain. This represents the most efficient structural improvement per unit of design change.

2. **The flat maple deck fails.** $\text{FoS} = 0.87 < 1$ under the 5,000 N design load. A two-foot drop by a 75 kg rider is sufficient to initiate bending fracture in a flat-profile maple deck. Real-world evidence supports this: flat boards are more prone to snapping than properly concave ones.
3. **CFRP provides a 48% deformation advantage** over the flat maple baseline, with an enormous bending FoS of 26.5. However, the CFRP laminate’s actual failure mode would be inter-ply delamination at truck interfaces, not fiber fracture. Local interface reinforcement (aluminum inserts, larger truck mounting washers, titanium hardware) would be the highest priority design improvement for a CFRP deck.
4. **The recommended design is the curved CFRP deck.** It satisfies all structural criteria with large margin, reduces midspan deflection by 48% versus the current industry-standard flat maple baseline, and offers superior fatigue resistance. The primary disadvantages, higher cost and brittle failure mode, are manageable through appropriate design margins and hardware improvements.
5. **Future work** should include a mesh-converged full assembly FEA at 10 to 15 mm element size, a nonlinear contact analysis with bolt preload at the truck zone, an explicit dynamic impact simulation using shell elements and symmetry, and fatigue life prediction using S-N data for CFRP laminates under cyclic bending.

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